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विश्वजीवनामृतं ज्ञानम्

Education

Degree	Institution / Board	CGPA / %	Year
Integrated B.Tech + M.Tech (IT)	Atal Bihari Vajpayee Indian Institute of Information Technology and Management, Gwalior	8.92 / 10	2023 – 2028
Class XII	CBSE	96.67%	2023
Class X	CBSE	95.8%	2021

Top 5 in batch | *JEE Main 99th percentile* | *MCM Scholarship* | *Reliance Foundation UG Scholar* | *Verified Gradesheets*

Experience

Open Source Developer (Google Summer of Code) **Aug 2025 – Present**

omegaUp: online judge & education platform (Vue, TypeScript, PHP 8.1, MySQL 8, Python) *Remote*

- Completed **Google Summer of Code 2026** (350 h, passed both evaluations) [Certificate] [Verification Letter] [Write-up]: built a **Cron Control Plane** for 5 production batch jobs, a shared Python runner with **MySQL advisory locks** blocking overlapping runs across hosts, per-phase telemetry and run history.
- Shipped an **admin dashboard** (Vue.js, TypeScript, PHP REST APIs) for job health and failures, with a **secure rerun path**: the API only queues requests and an allow-listed dispatcher executes them.
- Added **reliability guardrails**: in-transaction ranking checks that roll back anomalous results, publish gates that reject regressed ML models, and nightly health checks for 4 classes of broken problems.
- **60 merged PRs** (40 since May 2026, +14K lines) incl. GitHub OAuth sign-in; wrote **82 Python unit tests**; collaborated with 2 mentors through design and code reviews.

Project Intern **May 2024 – Aug 2024**

Advanced Materials Research Lab, ABV-IIITM *Gwalior, India*

- Engineered a multi-sensor **IoT device** (Arduino, C++) monitoring cattle temperature, heart rate and gas levels; streamed readings over Wi-Fi/Bluetooth to Firebase and Google Sheets with a mobile dashboard.

Projects

ASTRA-IDE: Self-Scheduling Cloud IDE *(Kubernetes, gVisor, eBPF, PyTorch)* May 2026 – Present |

- Deployed a scalable cloud IDE with per-workspace containers and risk-based isolation (runc, gVisor or microVM) across **Kubernetes clusters in 3 regions**; Yjs live collaboration; CI/CD via GitHub Actions.
- Cut cold starts **49.4%** with LSTM pre-warming and emissions **40.7%** with carbon-aware scheduling; syscall-graph intrusion detector reaches **F1 0.816** at 4% FPR (first-author paper under review).

Einstein: Local-First AI Contact Manager *(Next.js, FastAPI, CRDT, WebRTC)* Aug 2025 – Present |

- Hand-rolled a **CRDT** (hybrid logical clocks, OR-Set) for offline, conflict-free peer-to-peer sync over WebRTC; prompt-engineered LLM features with JSON outputs: natural-language search and business-card OCR.

ChurnAware: Sentiment-Aware Churn Prediction *(Python, scikit-learn, PyTorch)* Sep – Nov 2025 |

- GMM segmentation and a leakage-free temporal churn model with review sentiment on Blinkit e-grocery data, plus a Double-DQN engine recommending retention incentives.

Technical Skills

Languages: C++, Python, TypeScript, JavaScript, PHP, SQL, C, HTML/CSS
Backend & Frontend: REST APIs, FastAPI, Node.js, Express.js, React, Next.js, Vue.js, Tailwind CSS
Databases: MySQL, PostgreSQL, MongoDB, Redis
Cloud & DevOps: Docker, Kubernetes, GitHub Actions (CI/CD), Linux, Git, PHPUnit, Jest, Cypress
AI/ML: PyTorch, scikit-learn, Pandas, LLM APIs, prompt engineering, GitHub Copilot
Coursework: Data Structures & Algorithms, OOP, DBMS, Computer Networks, Software Engineering

Achievements

- **LeetCode:** max contest rating **1,868** (top 8.5%), 540+ problems solved.
- **Webkriti 2024 Winner:** 1st place in a 3-month full-stack web development contest at IIITM Gwalior.
- **GATE 2026 (CS):** qualified in 3rd year. [Scorecard]